

Chapter 7

Determinants

Only One Option Correct Type Questions

1. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form

$$\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$$

where each of a, b and c is either ω or ω^2 . Then the number of distinct matrices in the set S is

[IIT-JEE-2011 (Paper-2)]

- (A) 2 (B) 6
(C) 4 (D) 8

One or More Option(s) Correct Type Questions

2. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is (are)

[IIT-JEE-2012 (Paper-2)]

- (A) -2 (B) -1
(C) 1 (D) 2

3. Which of the following values of α satisfy the equation

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha?$$

[JEE (Adv)-2015 (Paper-1)]

- (A) -4 (B) 9
(C) -9 (D) 4

4. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$, where $k \in \mathbb{R}$, $k \neq 0$

and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then

[JEE (Adv)-2016 (Paper-1)]

- (A) $\alpha = 0, k = 8$ (B) $4\alpha - k + 8 = 0$
(C) $\det(P \operatorname{adj}(Q)) = 2^9$ (D) $\det(Q \operatorname{adj}(P)) = 2^{13}$

5. Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

Which of the following statement(s) is(are) correct?

[JEE (Adv)-2016 (Paper-2)]

- (A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ
- (B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ
- (C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$
- (D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$
6. For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

If Q is a nonsingular matrix of order 3×3 , then which of the following statements is(are) TRUE?

[JEE (Adv)-2021 (Paper-1)]

(A) $F = PEP$ and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$

(C) $|(EF)^3| > |EF|^2$

(D) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

Integer / Numerical Value Type Questions

7. Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations:

$$3x - y - z = 0$$

$$-3x + z = 0$$

$$-3x + 2y + z = 0$$

Then the number of such points for which $x^2 + y^2 + z^2 \leq 100$ is

[IIT-JEE-2009 (Paper-2)]

8. Let k be a positive real number and let $A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$.

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$, then $[k]$ is equal to

[IIT-JEE-2010 (Paper-2)]

[Note : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k]

9. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is [JEE (Adv)-2016 (Paper-1)]

Question Stem for Question No. 10

Question Stem

Let α , β and γ be real numbers such that the system of linear equations

$$x + 2y + 3z = \alpha$$

$$4x + 5y + 6z = \beta$$

$$7x + 8y + 9z = \gamma - 1$$

is consistent. Let $|M|$ represent the determinant of the matrix

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equations is consistent, and D be the **square** of the distance of the point $(0, 1, 0)$ from the plane P . [JEE (Adv)-2021 (Paper-1)]

10. The value of $|M|$ is _____.

Matrix-Match / Matrix Based Type Questions

11. Let p , q , r be non-zero real numbers that are, respectively, the 10^{th} , 100^{th} and 1000^{th} terms of a harmonic progression. Consider the system of linear equations [JEE (Adv)-2022 (Paper-1)]
- $$x + y + z = 1$$
- $$10x + 100y + 1000z = 0$$
- $$qr x + pr y + pq z = 0$$

	List-I		List-II
(I)	If $\frac{q}{r} = 10$, then the system of linear equations has	(P)	$x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution
(II)	If $\frac{p}{r} \neq 100$, then the system of linear equations has	(Q)	$x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as a solution
(III)	If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R)	infinitely many solutions
(IV)	If $\frac{p}{q} = 10$, then the system of linear equations has	(S)	no solution
		(T)	at least one solution

The **correct** option is:

- (A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T) (B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)
 (C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R) (D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

